WORKED SOLUTION

STEP I 1996, Question 5

Pure • Complex Square Roots

> **STEP I 1996, Question 5**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/96-s1-q5

The key idea

Each part feeds the next. Part (i) is about matching rational and irrational parts, and it tells us that $4-2\sqrt{3}$ is a perfect square. Part (ii) asks for a complex square root, and after the usual "compare real and imaginary parts" step it collapses onto part (i). Part (iii) is a quadratic whose discriminant turns out to be four times the number in part (ii), so the square root we need is already sitting there.

Solution

Part (i)

Expanding,

$$(r+s\sqrt{3})^2 = (r^2+3s^2)+2rs\sqrt{3}.$$

We want this to equal $4-2\sqrt{3}$. Since r and s are rational, so are r^2+3s^2 and $2rs$. If $a+b\sqrt{3}=c+d\sqrt{3}$ with a,b,c,d rational and $b\neq d$, then $\sqrt{3}=(a-c)/(d-b)$ would be rational, which it isn't. So we may compare the rational parts and the coefficients of $\sqrt{3}$:

$$r^2+3s^2=4, \qquad rs=-1.$$

Then $r\neq 0$, so $s=-1/r$, and the first equation becomes

$$r^2+\frac{3}{r^2}=4 \iff r^4-4r^2+3=0 \iff (r^2-1)(r^2-3)=0.$$

The option $r^2=3$ gives $r=\pm\sqrt{3}$, which is not rational. So $r^2=1$, and the solutions are

$$r=1,\ s=-1 \qquad \text{and} \qquad r=-1,\ s=1,$$

that is, $r+s\sqrt{3}=\pm(1-\sqrt{3})$. Indeed $(1-\sqrt{3})^2=1-2\sqrt{3}+3=4-2\sqrt{3}$.

Part (ii)

Expanding $(p+q\mathrm{i})^2 = (p^2-q^2)+2pq\,\mathrm{i}$. Since p and q are real we can compare real and imaginary parts:

$$p^2-q^2 = 3-2\sqrt{3}, \qquad pq = 1-\sqrt{3}.$$

The second equation shows $p \neq 0$, so $q = (1-\sqrt{3})/p$ and $q^2 = (1-\sqrt{3})^2/p^2 = (4-2\sqrt{3})/p^2$. Substituting into the first equation and multiplying by p^2,

$$p^4 - (3-2\sqrt{3})\,p^2 - (4-2\sqrt{3}) = 0.$$

Ask yourself This is a quadratic in p^2 with nasty-looking coefficients. Is there a quick way to factorise it?

The two roots for p^2 must have sum $3-2\sqrt{3}$ and product $-(4-2\sqrt{3})$. Try -1: the other root would then be $4-2\sqrt{3}$, and indeed $-1+(4-2\sqrt{3}) = 3-2\sqrt{3}$. So

$$(p^2+1)(p^2-(4-2\sqrt{3})) = 0.$$

Since p is real, $p^2 \neq -1$, so $p^2 = 4-2\sqrt{3} = (\sqrt{3}-1)^2$ by part (i). Hence $p = \pm(\sqrt{3}-1)$, and $q = (1-\sqrt{3})/p = \mp 1$. The solutions are

$$p = \sqrt{3}-1,\ q=-1 \qquad \text{and} \qquad p = 1-\sqrt{3},\ q = 1.$$

In other words, the square roots of $(3-2\sqrt{3})+2(1-\sqrt{3})\mathrm{i}$ are $\pm((\sqrt{3}-1)-\mathrm{i})$. As a check, $((\sqrt{3}-1)-\mathrm{i})^2 = (4-2\sqrt{3})-1-2(\sqrt{3}-1)\mathrm{i}$, which is the required number.

Part (iii)

By the quadratic formula (which holds for complex coefficients, because completing the square works just as well over $\mathbb{C}$),

$$z = \frac{2 \pm \sqrt{4-4(1+\mathrm{i})(2\sqrt{3}-2)}}{2(1+\mathrm{i})} = \frac{1 \pm \sqrt{1-(1+\mathrm{i})(2\sqrt{3}-2)}}{1+\mathrm{i}},$$

where $\pm\sqrt{\ }$ stands for the two complex square roots. The quantity under the root is

$$1-(2\sqrt{3}-2)-(2\sqrt{3}-2)\mathrm{i} = (3-2\sqrt{3})+2(1-\sqrt{3})\mathrm{i},$$

which is exactly the number in part (ii). Its square roots are $\pm((\sqrt{3}-1)-\mathrm{i})$, so

$$z = \frac{1 \pm ((\sqrt{3}-1)-\mathrm{i})}{1+\mathrm{i}}, \quad \text{giving} \quad z_1 = \frac{\sqrt{3}-\mathrm{i}}{1+\mathrm{i}}, \qquad z_2 = \frac{(2-\sqrt{3})+\mathrm{i}}{1+\mathrm{i}}.$$

Multiplying top and bottom by $1-\mathrm{i}$ (so the denominator becomes 2):

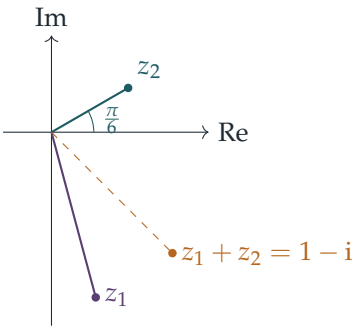
$$z_1 = \frac{(\sqrt{3}-\mathrm{i})(1-\mathrm{i})}{2} = \frac{(\sqrt{3}-1)-(\sqrt{3}+1)\mathrm{i}}{2},$$
$$z_2 = \frac{((2-\sqrt{3})+\mathrm{i})(1-\mathrm{i})}{2} = \frac{(3-\sqrt{3})+(\sqrt{3}-1)\mathrm{i}}{2}.$$

Ask yourself Is this "as simple a form as possible"? Can either root be tidied further?

The second root factorises: $(3-\sqrt{3})+(\sqrt{3}-1)\mathrm{i} = (\sqrt{3}-1)(\sqrt{3}+\mathrm{i})$, and $\sqrt{3}+\mathrm{i} = 2\mathrm{e}^{\mathrm{i}\pi/6}$. For the first, $|z_1|^2 = \frac{1}{4}((4-2\sqrt{3})+(4+2\sqrt{3})) = 2$, and $z_1 = \frac{\sqrt{3}-\mathrm{i}}{1+\mathrm{i}}$ is a quotient of $2\mathrm{e}^{-\mathrm{i}\pi/6}$ by $\sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$. So the solutions are

$$z_1 = \tfrac{1}{2}((\sqrt{3}-1)-(\sqrt{3}+1)\mathrm{i}) = \sqrt{2}\,\mathrm{e}^{-5\pi\mathrm{i}/12},$$
$$z_2 = \tfrac{1}{2}(\sqrt{3}-1)(\sqrt{3}+\mathrm{i}) = (\sqrt{3}-1)\,\mathrm{e}^{\pi\mathrm{i}/6}.$$

Either form (Cartesian or modulus–argument) is a perfectly good final answer.



Checks

- Sum and product of roots: for $az^2+bz+c=0$ we need $z_1+z_2 = -b/a = 2/(1+\mathrm{i}) = 1-\mathrm{i}$ and $z_1z_2 = c/a = (2\sqrt{3}-2)/(1+\mathrm{i}) = (\sqrt{3}-1)(1-\mathrm{i})$. From the Cartesian forms, $z_1+z_2 = \frac{1}{2}(2-2\mathrm{i}) = 1-\mathrm{i}$; from the polar forms, $z_1z_2 = \sqrt{2}(\sqrt{3}-1)\mathrm{e}^{-\mathrm{i}\pi/4} = (\sqrt{3}-1)(1-\mathrm{i})$. Both agree.
- Numerically, $z_1 \approx 0.366 - 1.366\mathrm{i}$ and $z_2 \approx 0.634 + 0.366\mathrm{i}$, and substituting either into $(1+\mathrm{i})z^2 - 2z + 2\sqrt{3} - 2$ gives 0 to machine precision.

Common slips

- In part (i), forgetting to say *why* we may compare coefficients of $\sqrt{3}$: it relies on r, s being rational and $\sqrt{3}$ irrational. Without it, there are infinitely many real solutions.
- In part (ii), keeping $p^2 = -1$, or pairing the signs of p and q wrongly: $pq = 1-\sqrt{3} < 0$ forces opposite signs.
- In part (iii), dividing the whole formula by 2 incorrectly, or leaving $1+\mathrm{i}$ in the denominator rather than realising it.

Beyond STEP

Part (i) works because 1 and $\sqrt{3}$ are linearly independent over $\mathbb{Q}$, so $\mathbb{Q}(\sqrt{3}) = \{r + s\sqrt{3}\}$ behaves like a two-dimensional space over the rationals; part (ii) uses the same idea with 1 and i over $\mathbb{R}$. This is the start of field extensions, which a first-year algebra course develops towards Galois theory. Allenby's *Rings, Fields and Groups* covers this, and Needham's *Visual Complex Analysis* gives a beautiful geometric view of complex square roots.

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